Your Starter Pack to Agentic Automation [Tamil Edition]

Autonomous Agent — Use Case & Problem Statement

AI Stock Price Lookup Agent | API-Workflow Domain (Investment Desk)

Detail	Description
Audience	Engineering & Arts students (challenge exercise)
Agent Type	Autonomous Agent (no human-in-the-loop)
Platform	UiPath Autopilot / Agent Builder
Difficulty	Moderate — introduces live external stock market API integration

1. Problem Statement

Investment desk employees and finance analysts frequently email each other or a shared inbox asking for the latest price of a stock — for example, checking a listed company's current share price before a portfolio review meeting, or confirming a ticker's movement before an internal report is finalized. Today, staff manually open a market data terminal or financial website, look up the price, and reply by email — a small but repetitive task that interrupts focused analysis work and does not scale when many such requests arrive during market hours.

Design and build an Autonomous UiPath Agent, the AI Stock Price Lookup Agent, that reads an incoming email requesting a stock price, extracts the ticker symbol or company name, calls a live Stock Market API workflow to fetch the current price and day's change, and independently replies with the result — without any human approval step.

2. System Prompt

```
You are an AI Stock Price Lookup Agent for the Investment Desk. Your goal
is to analyze stock price request emails and provide an accurate,
up-to-date market price using a live stock market API.

Follow this reasoning process:
1. Read and understand the email content.
2. Classify it into one of the following:
   - Stock Price Request
   - Price Movement / Alert Enquiry
   - Other
3. If it is Stock Price Request or Price Movement/Alert Enquiry:
   - Extract the ticker symbol or company name and the exchange
     (if mentioned)
   - Call the Stock Market API workflow with the ticker symbol
   - Decide if it is a notable movement (e.g., absolute change greater
     than 5% in a single day)
   - Generate a professional reply stating the current price, the
     day's change (value and percentage), and the price timestamp
4. If it is Other:
   - Skip processing
```

```
Rules:
- Think step-by-step before answering
- Be concise and structured
- Do not assume missing details; if the ticker symbol or company name
  is unclear, ask the requester to clarify instead of guessing
- Do not invent price values — always use the API response
- Return output in structured format
```

3. User Prompt

```
Analyze the following stock price request email and provide the current
market price using the live stock market API:
{{stock_email_text}}
```

4. Input Schema

```
{
  "stock_email_text": "string"
}
```

5. Output Schema

```
{
  "category": "string",
  "ticker_symbol": "string",
  "company_name": "string",
  "current_price": "string",
  "day_change_value": "string",
  "day_change_percent": "string",
  "price_timestamp": "string",
  "notable_movement": "true | false",
  "action": "string"
}
```

6. Tool: API Workflow & Publish

Create the following process inputs for the API workflow (HTTP Request activity) the agent will invoke:

- ticker_symbol (text) — e.g., "TCS", "INFY"
- exchange (text) — optional, e.g., "NSE" or "BSE"

Configure the workflow to call a live stock market API endpoint (e.g., an HTTPS GET request to a market-data provider) and parse the JSON response for current price, day's change, and last-updated timestamp.

7. Add Tool: Map Inputs

Agent Field	API Request Parameter
ticker_symbol	symbol
exchange	market_code

8. Updated System Prompt — Tool Invocation Logic

```
If the email is classified as Stock Price Request or Price
Movement/Alert Enquiry:
  - Call the get_stock_price API workflow with ticker_symbol and
    exchange (if available).
  - Parse the API response for 'current_price', 'day_change_value',
    'day_change_percent', and 'last_updated'.
  - If the absolute value of day_change_percent exceeds 5%, set
    notable_movement = "true".
  - Use the generate_reply GenAI activity to draft a professional reply
    email with the current price, day's change, and timestamp — fully
    autonomously, with no manual approval required.

If the email is classified as Other:
  - Take no further action.
```

9. Tools & Activities Summary

Tool / Activity	Type
get_stock_price	API Workflow (HTTP Request to a live stock market data API)
generate_reply	GenAI Activity (LLM prompt-based text generation)
classify_request	GenAI Activity (embedded in Agent reasoning)

Note: This use case is intentionally fully autonomous — the agent classifies the request, calls the live market data API, evaluates price movement, and composes the reply without an escalation or human-review step, demonstrating end-to-end autonomous decision-making for the workshop challenge.

10. Expected Learning Outcome

- Design a system prompt that integrates a live external market data API into agent reasoning
- Configure an API Workflow (HTTP Request activity) with mapped input parameters
- Parse a JSON API response and derive a notable-movement flag from the returned data
- Define structured input/output schemas for an API-integrated autonomous agent
- Publish and test a fully autonomous, API-driven agent in UiPath Agent Builder